

Technology Stack

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Overview

Smart Lender uses a lightweight, Python-centric stack suitable for rapid ML prototyping and deployment as a simple web application.

Languages

- Python 3.x — core language for data processing, ML, and backend
- HTML/CSS — frontend templates

Data Science & ML Libraries

Library	Purpose
Pandas / NumPy	Data loading, cleaning, manipulation
Matplotlib / Seaborn	Exploratory data analysis & visualization
Scikit-learn	Decision Tree, Random Forest, KNN, preprocessing, metrics
XGBoost	Gradient boosting classifier
imbalanced-learn (SMOTE)	Balancing the dataset

Application Layer

Component	Technology
Backend	Flask
Frontend	HTML, CSS (Jinja2 templates)
Model Serialization	Pickle (.pkl)

Tooling

- Jupyter Notebook / Google Colab — for EDA and experimentation
- Git & GitHub — version control
- VS Code — primary IDE